

Curtis Ho, Undergraduate at The University of Toronto

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Portfolio - <https://curtisho3.github.io/portfolio/>

SKILLS SUMMARY

Budding Mechanical Engineer with an ever-evolving ambition in revolutionizing the transportation and energy sector.

- Mechanical: SOLIDWORKS and CATIA for CAD **Modeling and Rendering**, ANSYS for Multi-force **FEA Analyses**, Blender **3D Animations**
- Electrical and Programming: Arduino for **IoT**, Python PyTorch for Training **Deep Learning** Models, MATLAB and R for **Data Analysis**, C for other **Programming**
- Machining: Lathe, Mill, Drill Press
- Others: QGIS, LLM Prompt Engineering, 3D Printing with Prusa and Ultimaker
- Soft Skills: Team Leadership, Adaptability, Curriculum Development

EDUCATION

BASc in Mechanical Engineering + PEY Co-op 2024 – 2029
The University of Toronto, Toronto, Ontario
Intended Minors: Environmental Engineering and Robotics and Mechatronics

High School Diploma 2020 – 2024
Northfield Mount Hermon School, One Lamplighter Way, Gill, Massachusetts
Dean's List

AWARDS AND ACHIEVEMENTS

- MIT Health Hack Aging-in-Place Winning Team 2021
- NMH 3D Printed Car Derby Race Third Runner Up 2021
- The Cum Laude Society 2023

Certifications

- Elementary Level of the International Tree Climbing Association
- Completion of Udemy's 100 Days of Coding Course
- SolidWorks Design Associate (CSWA)

Volunteer Work

Ecological Researcher, Costa Rica Wildlife & Environmental Conservation Program 2023

- Conducted field survey on dolphin distribution, bamboo health, and Macaw population
- Studied locals' environment conservation techniques

Community Service in Shaoguan, China 2019

- Visited families of underprivileged high school students in Shaoguan, China to understand and report their needs

EXPERIENCE

University of Toronto Underwater Exploration Design Team Oct - 2025 to Present

Mechanical Lead

- Oversee design for Niskin samplers, buoyancy controls, nose cones, and other core components for an autonomous underwater drone
- Engineered a custom waterproof inner-hull housing critical power and PCB systems

Common Farms

Jun - 2026 to Jul - 2036

Engineering Intern – Agricultural Systems

- Automated climate-based irrigation for 30 vertical racks by integrating Tuya IoT sensors with Netro controllers across a 3,000 sq ft facility
- Rapid-prototyped custom 3D-printed hardware (LED mounts, protective port covers) to optimize light distribution and prevent pest infestation
- Co-led hands-on agricultural curricula for a summer camp on sustainable farming and hydroponics for youth age groups (3–11 years old)

ReGen Technology Holdings, Limited

Apr - 2025 to Aug - 2025

Engineering and Marketing Intern

- Pioneered a solar-thermal desalination system that produces nutrient enriched irrigation ready fresh water
- Developed AI agentic workflow for automated email outreach to potential customers, maintaining a 1000+ customer pipeline with minimal manual oversight
- Prototyped adjustable tilt-control mechanism for calibrating silos on uneven terrain

STEMpire Bridge

Jun - 2023 to Jun - 2024

Cofounder

- *Established* a STEM tutoring *social-enterprise* developing Python and 3D modeling curricula to 30+ students across Massachusetts' public schools and charter schools

Solar Man Air Conditioning, Limited

Jun - 2022 to Aug - 2023

Research Assistant

- Utilized Blender for promotional 3D animations that secured funding/interest for a patented, refrigerant-free personal micro-turbine A/C
- Designed a solar desalination tower with embedded energy generation
- Executed digital growth, building company website and scaling YouTube channel to 1.8k+ subscribers and 600+ average likes.

MIT Hong Kong Innovation Node

Jun - 2023 to Aug - 2023

Youth Fellow

- Led a team of 5 high school students in the creation of several AI projects
- Integrated clinical insights from MDs to build a medical consultant chatbot with expertise in screening and prevention of cancer and cardiovascular diseases; achieved query accuracy of 97% through iterative prompt engineering
- Taught middle-schoolers at St. Paul's Coed Summer Portal, supported the instruction of MIT App Inventor for a project on enhancing the HK tram riding experience